MIL-C-27022A(USAF)
10 August 1960
Superseding
MIL-C-27022(USAF)
7 June 1960

MILITARY SPECIFICATION

CRADLE, BOMB HANDLING MHU-27A/E

1. SCOPE.- This specification establishes the requirements for one type of bomb cradle, designated as MHU-27A/E, hereafter referred to as the cradle.

2. APPLICABLE DOCUMENTS.- The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

UU-P-271	Paper, Wrapping, Water-proof Kraft
PPP-B-601	Boxes, Wood Cleated-Plywood
PPP-B-621	Boxes, Wood Nailed and Lock-Corner

Military

JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods Of
MIL-A-140	Adhesive Water-Resistant, Waterproof-Barrier Material
MIL-E-4970	Environmental Testing, Ground Support Equipment, General Specification For
MIL-T-5021	Tests, Aircraft Welding Operator's Certification
MIL-W-6858	Welding: Aluminum, Magnesium, Non-Hardening Steels or Alloys, and Titanium, Spot, Seam, and Stitch
MIL-T-9107	Test Reports, Preparation Of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

FSC 1190

MWM CORP. DAYTON, OHIO
9/60 1500

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DRAWINGS

Air Force

59E45093 Cradle, Bomb Handling MHU-27A/E

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provisions for preproduction testing.

3.2 CONSTRUCTION.- Materials, finishes, and construction details not otherwise covered by this specification shall be in accordance with drawing 59E45093.

3.2.1 WELDING.- All welding shall be in accordance with MIL-W-6858. During fabrication of the cradle, all welding shall be inspected by personnel who have been certified in accordance with MIL-T-5021.

3.3 PERFORMANCE

3.3.1 FUNCTIONAL.- The cradle shall be capable of the following functions after being subjected to the proof load and to the environments required by this specification:

a. Rotation of the supported store about its longitudinal axis.

b. Adjustment of the sway braces, using a standard open end wrench.

c. Adjustment of the forklift clamp bolts, using a standard open end wrench.

3.3.2 PROOF LOAD.- The cradle shall be capable of supporting a downward load of 4000 pounds, applied midway between the two cradle rockers, without permanent deformation of any member. It shall be capable of withstanding this load when resting on its four legs on a level, hard surface and when supported by a forklift beneath its base assembly. In addition, it shall withstand this load with the cradle rockers rotated to their limits from bottom dead center.

3.3.3 ENVIRONMENTAL.- The cradle shall be capable of satisfactory operation after subjection to the following environments as described in section 4 of this specification:

a. High Temperature
b. Low Temperature
c. Salt Spray

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3.3.4 COMPATIBILITY.- The cradle shall be compatible with special stores for which it is intended. It shall also be compatible with the MHU-14/C and MHU-20/C Clip-In Subassemblies when they are mounted on the MHU-19/E Cradle.

3.4 IDENTIFICATION OF PRODUCT.- All parts and assemblies shall be marked for identification in accordance with MIL-STD-130.

3.5 GOVERNMENT LOANED PROPERTY.- The following items of equipment shall be loaned to the contractor, upon his request, as Government loaned property, for the purpose of conducting the functional test of this specification:

Item No.	Stock No.	Item Description	Quantity
1	1115-652-0075	Bomb, Practice, Profile BDU-3/B	1
or			
1	1115-604-4233	Bomb, Loading Practice, Type MD-9	1

3.6 WORKMANSHIP.- The cradle, including all parts and accessories, shall be fabricated and finished in a manner such that criteria of appearance, fit, and adherence to specified tolerances shall be observed. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, plating, painting, machine screw assemblage, welding, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS.- The inspection and testing of the cradle shall be as follows:

a. Preproduction Tests. . . . . .See 4.2
b. Acceptance Tests . . . . . .See 4.3

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES.- The preproduction test samples shall consist of cradles representative of the production equipment.

4.2.2 DESIGNATED LABORATORY.- With the exception of the fit tests, preproduction tests shall be conducted at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.2.1 POINT OF INSPECTION FOR FIT TESTS.- The fit tests required by this specification shall be conducted at the Air Force Special Weapons Center, Kirtland Air Force Base, New Mexico.

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4.2.2.2 PREPRODUCTION TEST REPORT.- In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.2.3 PREPRODUCTION TESTS.- Preproduction tests shall consist of all tests described under "4.4 TEST METHODS" of this specification.

4.3 ACCEPTANCE TESTS.- Acceptance tests shall consist of the following individual tests as described under "4.4 TEST METHODS" of this specification:

a. Examination of Product
b. Conformance to Drawings
c. Operational Test, No Load

4.3.1 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 TEST METHODS

4.4.1 EXAMINATION OF PRODUCT.- Each cradle shall be given a complete visual examination to determine compliance with the requirements of this specification, including materials and workmanship.

4.4.2 CONFORMANCE TO DRAWINGS.- Each cradle shall be inspected for strict conformance to drawings and specifications and standards referenced therein.

4.4.3 FUNCTIONAL TEST.- While supporting a 2000 pound dummy load, (either the BDU-3/B Profile Practice Bomb or the Type MD-9 Loading Practice Bomb may be used), the following functions shall be performed:

a. Rotation of the supported store about its longitudinal axis.

b Adjustment of the sway braces, using a standard open end wrench.

c. Adjustment of the forklift clamp bolts, using a standard open end wrench.

Failure of the cradle to function properly under load shall be cause for rejection.

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4.4.4 OPERATIONAL TEST, NO LOAD.- This test procedure shall be identical to the functional test procedure, except that there shall be no load on the cradle. Failure of the cradle to function properly shall be cause for rejection.

4.4.5 PROOF LOAD TESTS.- The cradle shall be completely assembled and resting on a level, hard surface with both rockers at bottom dead center. Using a jig to simulate a cylindrical shape approximately 20 inches in diameter, with the sway braces adjusted, a 4000 pound load shall be applied downward through the center of the cradle. Using the same jig, the cradle shall be raised by a fork-lift, and the rockers rotated to their limits from bottom dead center. The 4000 pound load shall again be applied. No permanent deformation of any member shall occur. At the completion of this procedure, the cradle shall be subjected to the functional test. Failure to pass the functional test shall be considered as failure to pass the proof load tests.

4.4.6 HIGH TEMPERATURE TEST.- The high temperature test shall be in accordance with procedure II of MIL-E-4970. At the end of the 50 hour exposure period, and while still at 51.7°C (125°F), the cradle shall be subjected to the functional test. Failure to pass the functional test shall be considered as failure to pass the high temperature test.

4.4.7 LOW TEMPERATURE TEST.- The low temperature test shall be in accordance with procedure III of MIL-E-4970. At the conclusion of the exposure period, and while stabilized at 0°C (32°F), the cradle shall be subjected to the functional test. Failure to pass the functional test shall be considered as failure to pass the low temperature test.

4.4.8 SALT SPRAY TEST.- The salt spray test shall be in accordance with MIL-E-4970. At the conclusion of the exposure period, the cradle shall be subjected to the functional test. Failure to pass the functional test shall be considered as failure to pass the salt spray test.

4.4.9 FIT TESTS.- The cradle shall be subjected to the following fit tests:

a. Between the cradle and applicable special stores.

b. Between the cradle and standard fork-lift of one to two ton capacity.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND INTERIOR PACKAGING, LEVELS A AND C

5.1.1 PRESERVATION.- Unless otherwise specified, parts of the cradle not protected from corrosion by painting; plating or chemical treatment shall be preserved in accordance with MIL-C-116.

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5.1.2 UNIT PACKAGING.- The cradle shall be packed in a suitable interior container, and shall be securely fastened, cushioned, blocked, and braced within the container in accordance with JAN-P-100. The cradle shall be completely assembled with all attaching fittings and fins in their proper places with the following exceptions. The crank will be mounted in reversed position to reduce cubage. The cradle shall be rotated to smallest amount of cubage.

5.2 PACKING

5.2.1 LEVEL A AND B.- A number of unit packages designated by the procuring activity shall be packed in an exterior container in accordance with PPP-B-601 or PPP-B-621 with materials, nails and nail spacing as specified for type 3 load. The top, sides, and ends of the cradle shall be protected by water proof barrier material conforming to UU-P-271 and sealed with adhesive conforming to MIL-A-140.

5.2.2 LEVEL C.- The cradle shall be packed in a substantial wood shipping container constructed to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery. The container shall conform to the requirements of consolidated freight classification rules in effect at the time of shipment. The container shall be capable of withstanding storage, rehandling, and reshipment without the necessity of repacking.

5.3 MARKING OF SHIPMENT.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

CRADLE, BOMB HANDLING MHU-27A/E

6. NOTES

6.1 INTENDED USE.- The MHU-27A/E Bomb Handling Cradle is intended for use in loading applicable stores in the clip-in multiple store subassemblies. This cradle will support a single store and will be lifted by a small fork-lift (capacity between 1 and 2 tons). It will provide for rotation of the store about its longitudinal axis. It will be compatible with all the applicable stores and will load the stores into both the MHU-14/C and MHU-20/C Clip-In Subassemblies while they are on the MHU-19/E Cradle. The cradle will also be capable of standing on a level surface in such a manner that a store can be loaded onto it and then the entire assembly be picked up from the ground by a fork-lift.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

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b. Conditions under which Government loaned property will be made available to the contractor.

c. Required levels of preservation, packaging and packing.

d. Whether preproduction tests will be conducted at a laboratory designated by the procuring activity or by the contractor under the supervision of the procuring activity.

NOTE: Fit tests may be conducted at the Air Force Special Weapons Center. (See 4.2.2.1).

6.2.1 PREPRODUCTION TEST SAMPLES.- It is expected that the contract or purchase order will specify that a minimum of one cradle representative of the production cradle will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order should specify the point of inspection for these tests.

6.3 DEFINITIONS

a. PREPRODUCTION TESTS.- Preproduction tests are those tests made on cradles which have been submitted by a contractor on a production contract or purchase order, to prove, prior to his starting production, that the equipment meets the requirements of this specification. Any production prior to the approval of the preproduction articles shall be at the contractor's risk. (See 4.2).

b. ACCEPTANCE TESTS.- Acceptance tests are those tests accomplished on equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification and to determine that the units are equivalent in performance and construction to the approved preproduction samples. (See 4.3).

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacturer, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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